

CS & IT ENGINEERING



Operating System

Multithreading

One Shot



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Recap of Previous Lecture



Topic

Round Robin Scheduling

Topic

Multilevel Queue Scheduling

Topic

Multilevel Feedback Queue Scheduling

Topics to be Covered



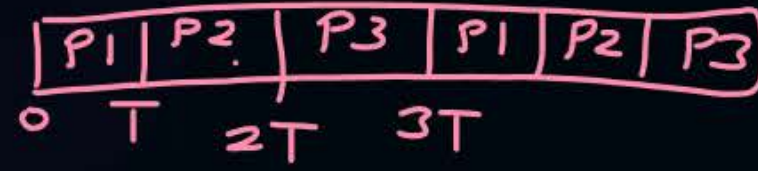
Topic

Multithreading

Topic

System Call: Fork()

[MCQ]



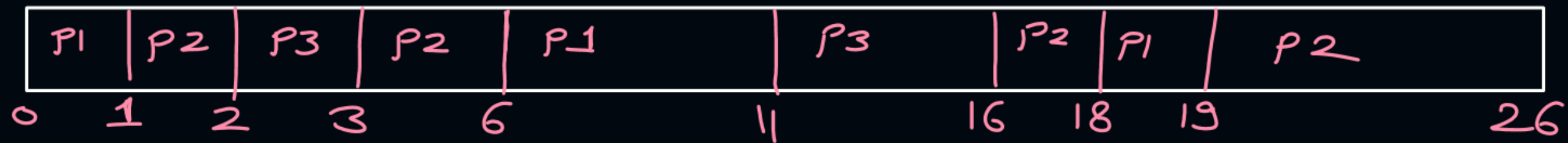
#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

GATE-2013

- A** This algorithm is equivalent to the first-come-first-serve algorithm
- B** ✓ This algorithm is equivalent to the round-robin algorithm. *with $Q = T$*
- C** This algorithm is equivalent to the shortest-job-first algorithm
- D** This algorithm is equivalent to the shortest-remaining-time-first algorithm

#Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses preemptive SRTF algorithm. Consider system has enough number of resources to carry out IO operations for all processes in parallel at a time. What is the average waiting time for the execution for the processes?

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time	CT	TAT	WT
P1	0	6	7	1	19	19	5
P2	1	4	2	9	26	25	10
P3	2	1	6	5	16	14	2
					$\text{avg WT} = \frac{17}{3} = 5.67$		



$$\begin{array}{l} \text{P3 I/O} \\ \hline 3 + 6 = 9 \end{array} \rightarrow$$

$$\begin{array}{l} \text{P2 I/O} \\ \hline 6 + 2 = 8 \end{array} \rightarrow$$

$$\begin{array}{l} \text{P1 I/O} \\ \hline 11 + 7 = 18 \end{array} \rightarrow$$

Ans = 10.5

- #Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses non preemptive SJF algorithm. Consider system has only those number of resources to carry out IO operations for atmost 2 processes in parallel at a time. What is the average waiting time [✓]for the execution for the processes?
in ready queue

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time
P1	0	5	4	5
P2	0	3	7	4
P3	0	2	10	7
P4	0	6	6	1



$$\begin{array}{|l} P_3 \text{ I/O} \\ \hline 2 + 10 = 12 \end{array} \rightarrow$$

$$\begin{array}{|l} P_2 \text{ I/O} \\ \hline 5 + 7 = 12 \end{array} \rightarrow$$

$$\begin{array}{|l} 10 + 2 \\ \hline \text{I/O} \\ \text{wait} \\ \text{for} \\ P_1 \end{array} \rightarrow \begin{array}{|l} 12 + 4 = 16 \\ \hline P_1 \\ \text{I/O} \end{array}$$

$\swarrow$
P1 waits for I/O

$$\begin{array}{|l} P_4 \text{ I/O} \\ \hline 16 + 6 = 22 \end{array} \rightarrow$$

	CT	TAT	WT with I/O	WT w/o I/O
P1	25	25	11	9
P2	20	20	6	6
P3	33	33	14	14
P4	26	26	13	13

$$\text{avg WT} = \frac{42}{4} = 10.5$$

[MCQ]

#Q. The arrival time, priority and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I.O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	AT	Priority	BT (CPU)	BT(I/O)	BT(CPU)
P1	0	2	1	5	3
P2	2	3(lowest)	3	3	1
P3	3	1(highest)	2	3	1

The multi – programmed operating system uses preemptive priority scheduling. What are the finish times of the process P1, P2 and P3 ? **GATE - 2006**



A 11, 15, 9



C 11, 16, 10



B ✓ 10, 15, 9



D 12, 17, 11



Topic : CPU Utilization



1. Without IO Operations :-

$$\text{CPU utilization} = 1.0 = 100\%$$

2. With IO Operations if a process need I/O for p fraction of time

$$\text{for } n=1 \text{ process, CPU utilization} = 1 - p$$

$$\text{for } n=2 \text{ } \text{---||---} = 1 - p^2$$

$$\text{for } n \text{ } \text{---||---} = \boxed{1 - p^n}$$

#Q. A computer system has degree of multiprogramming 10. And maximum I/O wait, that can be tolerated is 0.5. The CPU utilization is 99.9%?

$$= 1 - (0.5)^{10}$$

$$= 0.9999$$

$$= 99.9\%$$

#Q. Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst Time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

Gate - 2019

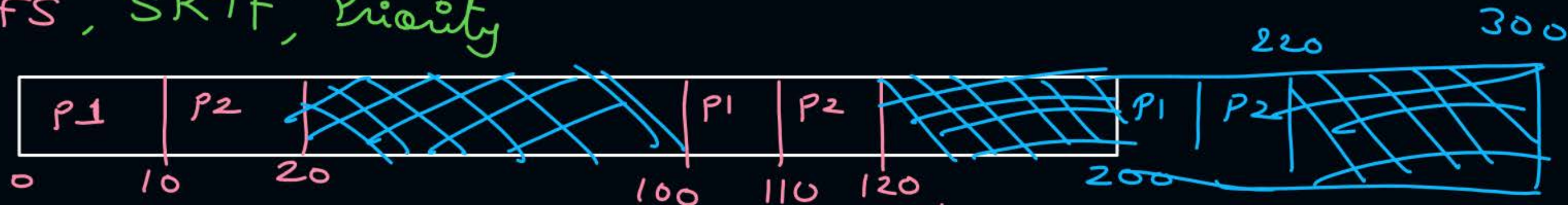
[MCQ]

#Q. A uni-processor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the process were created at nearly the same time. The I/O of both processes can proceed in parallel. Which of the following scheduling strategies will result in the least CPU utilization (over a long period of time) for this system ?

GATE - 2003

- A** First come first served scheduling
- B** Shortest remaining time first scheduling
- C** Static priority scheduling with different priorities for the two processes
- D** ✓ Round robin scheduling with a time quantum of 5 ms.

FCFS, SRTF, Priority



$$\left| \frac{P1 \text{ I/O}}{10 + 90 = 100} \right. \rightarrow$$

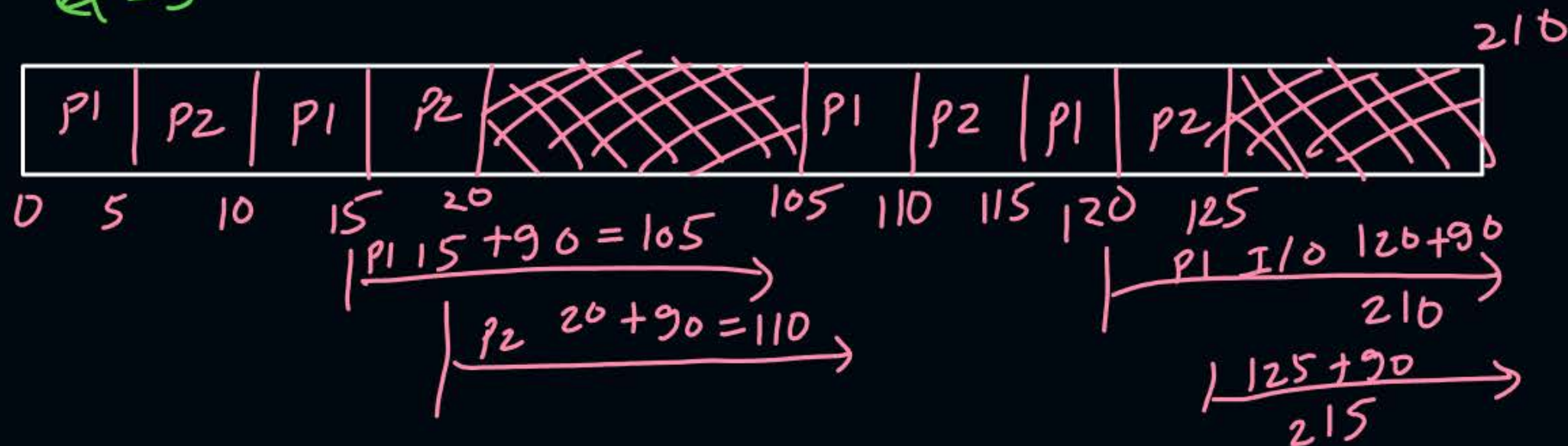
$$\frac{P1 \text{ I/O}}{110 + 90 = 200} \rightarrow$$

$$\left| \frac{P2 \text{ I/O}}{20 + 90 = 110} \right. \rightarrow$$

$$\left| \frac{P2 \text{ I/O}}{120 + 90 = 210} \right. \rightarrow$$

$$\text{CPU utilization} = \frac{20}{100} = 0.2$$

RR:- $Q=5$



$$\text{CPU utilization} = \frac{20}{105} = 0.19$$

H.W.

#Q. Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process gets blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

GATE - 2006**A**

0%

B

10.6%

C

30.0%

D

89.4%

#Q. Consider four processes P, Q, R, and S scheduled on a CPU as per round-robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

GATE-2022

- (A) $P = 4, Q = 10, R = 6, S = 2$
- (B) $P = 2, Q = 9, R = 5, S = 1$
- (C) $P = 4, Q = 12, R = 5, S = 4$
- (D) $P = 3, Q = 7, R = 7, S = 3$



Topic : Multithreading



Component of process
or
Lightweight Process

each thread can execute independently

Provide a way to improve application performance through parallelism



example:-

```
Sum = 0;
for (i = 1; i <= 100000; i++)
{
    Sum = Sum + i;
}
```

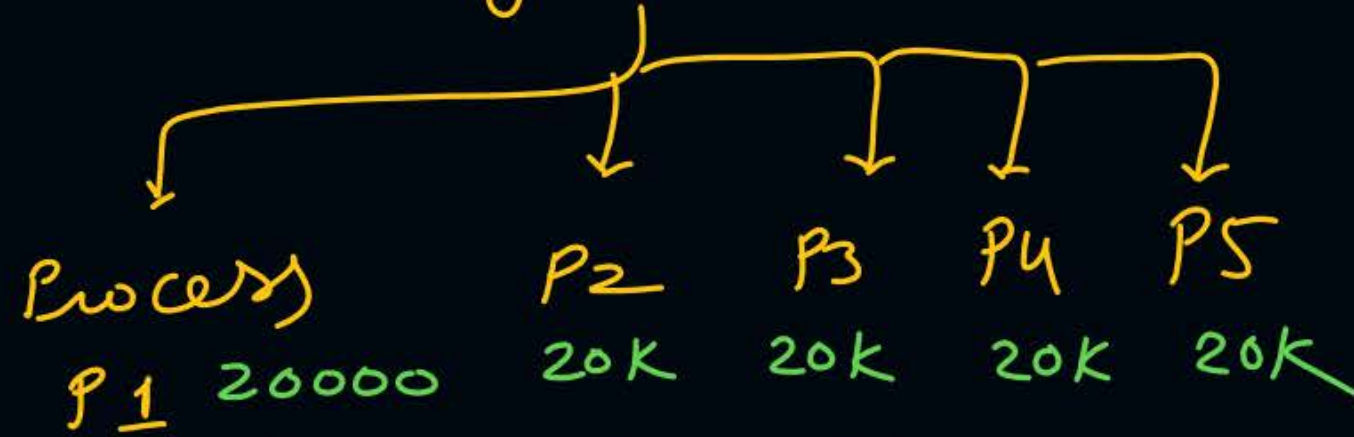
sum of first
1 Lac natural
numbers

↓
single process }
single CPU }

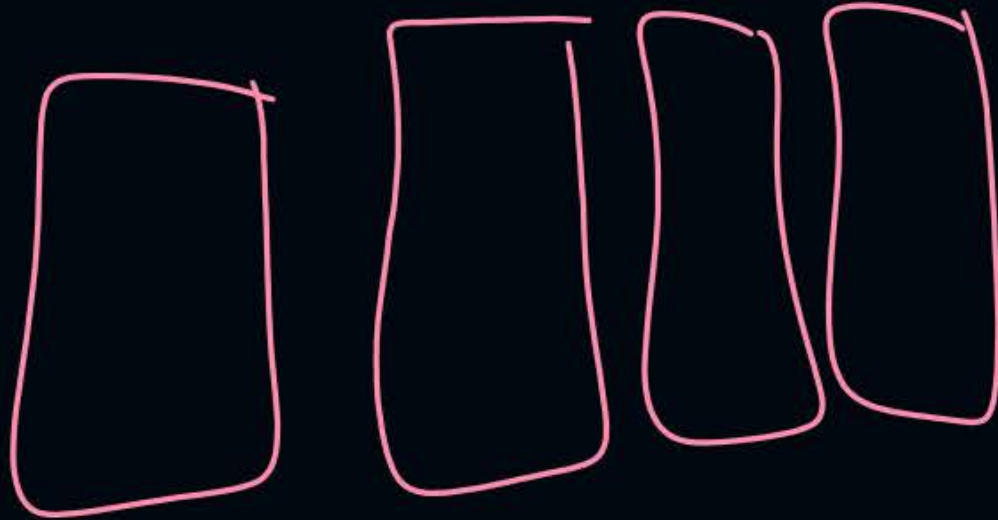
A lot of time
needed to execute

→ single process } single threaded process
5 CPUs }
only one CPU used to run entire process.

Single process

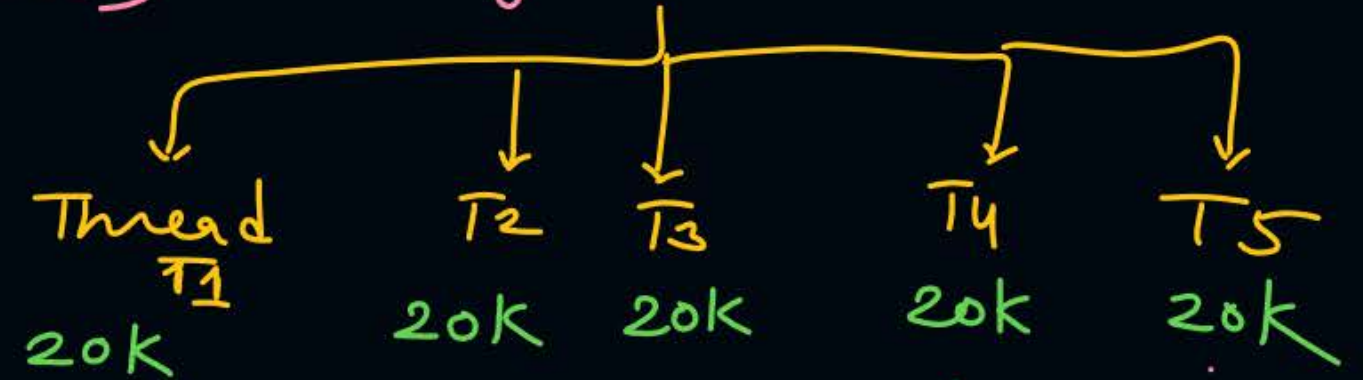


make 5 process



Solⁿ

Single process





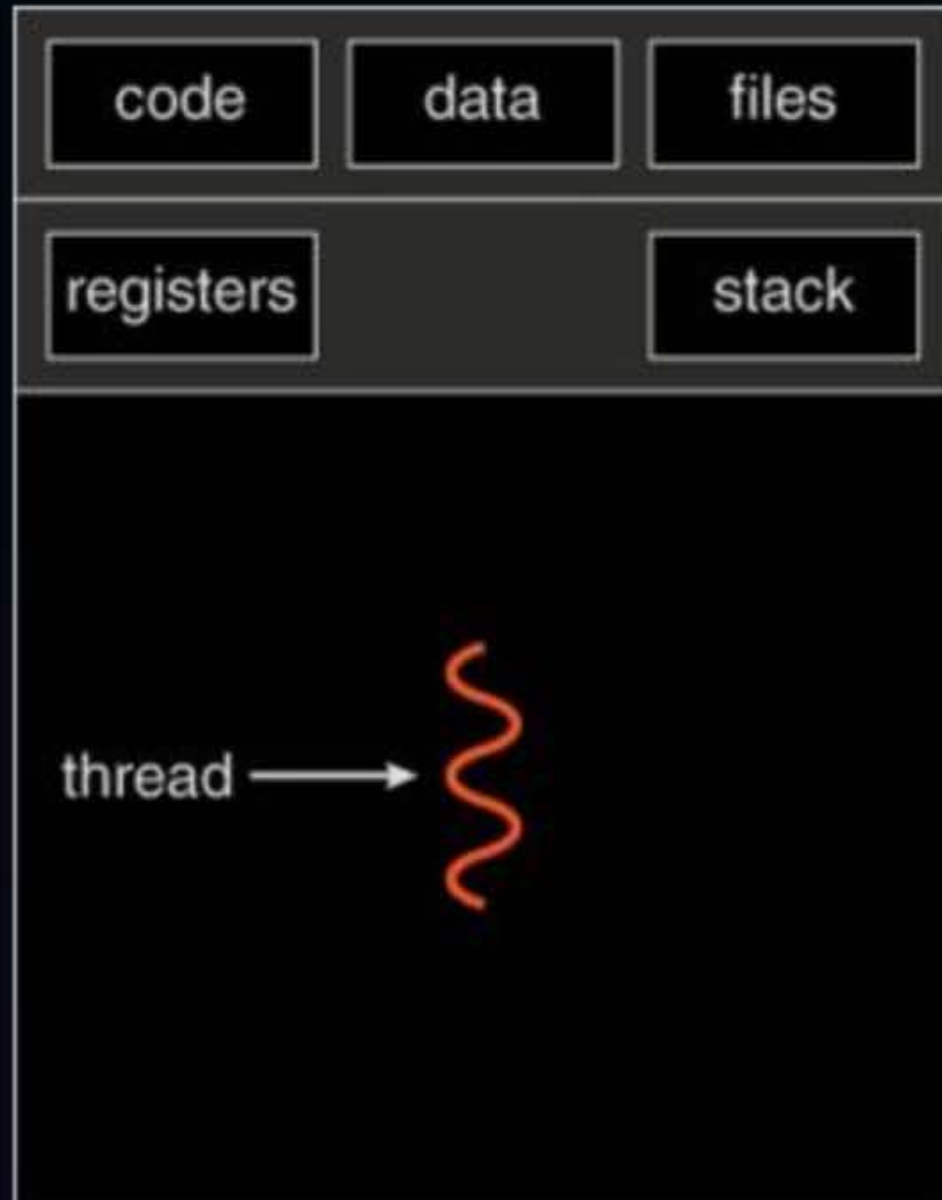
Topic : Threads



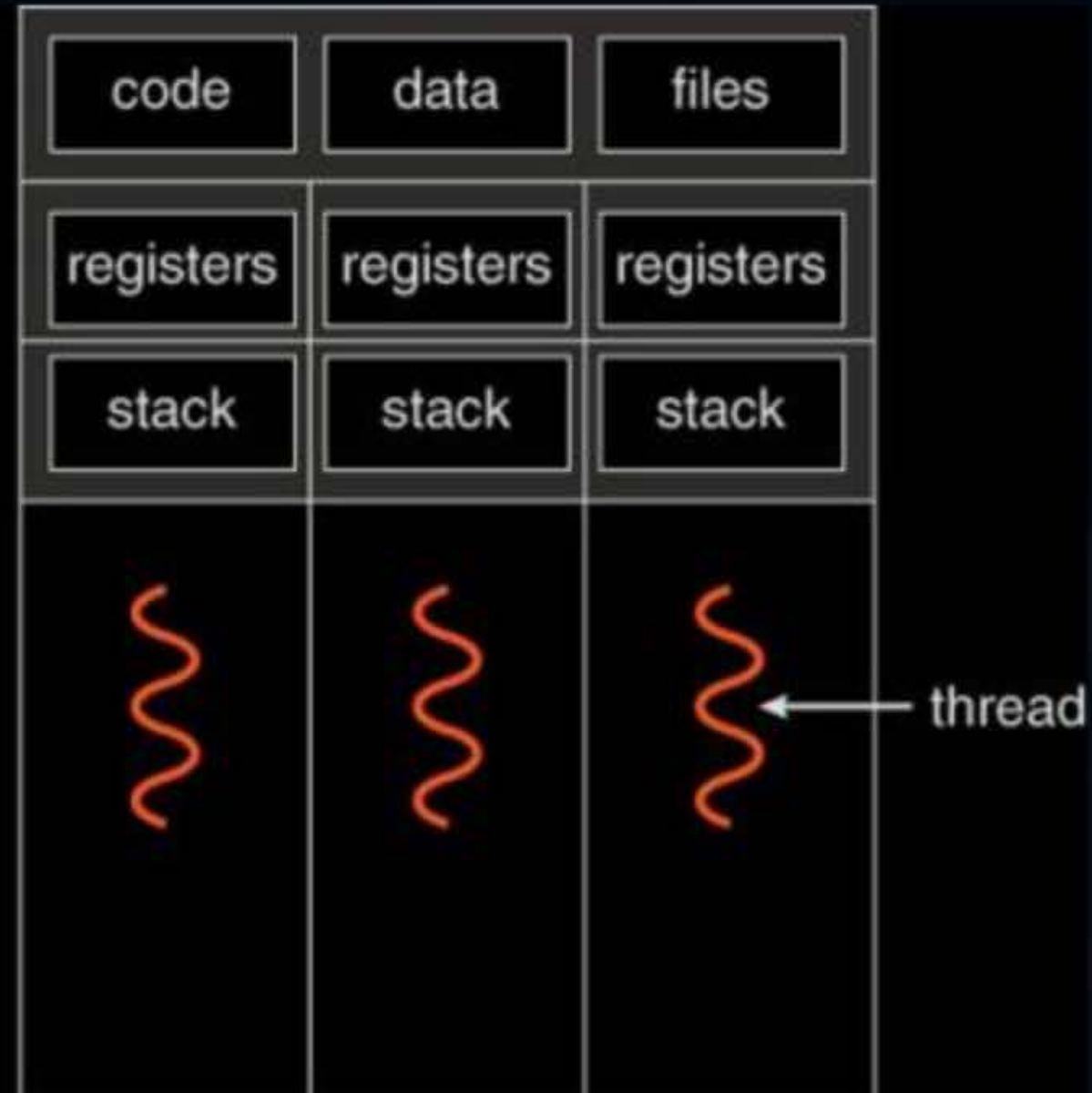
Shared Among Threads	Unique For Each Thread
Code Section	Thread Id
Data Section	Register Set
OS Resources	Stack
Open Files & Signals	Program Counter
Heap	



Topic : Threads



single-threaded process

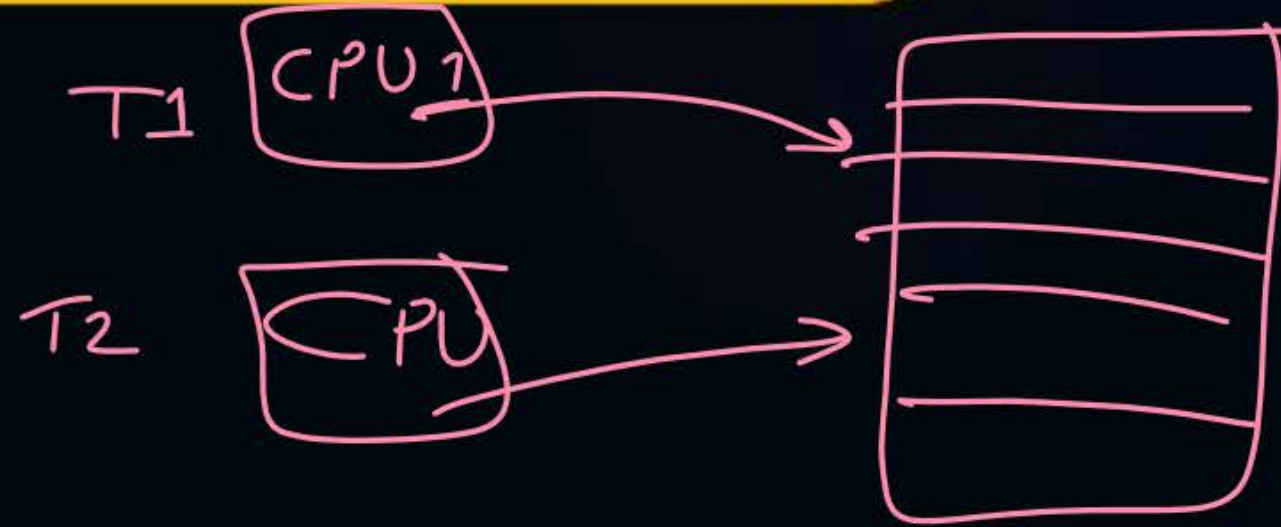


multithreaded process



Topic : Advantages of Multithreading

- Responsiveness
- Faster Context Switch
- Resource Sharing
- Economy
- Communication
- Utilization of Multiprocessor Architecture



Threads do not require OS for communication



Topic : Types of Threads



1. User Level Thread
2. Kernel Level Thread



Topic : Types of Threads

User Threads	Kernel Thread
Multithreading in user process	Multithreading in kernel process
Created without kernel intervention	Kernel itself is multithreaded
Context switch is very fast	Context switch is slow
If one thread is <u>blocked</u> , OS blocks entire process <i>for I/O</i>	Individual thread can be blocked
Generic and can run on any OS	Specific to OS
Faster to create and manage	Slower to create and manage



Topic : System Call

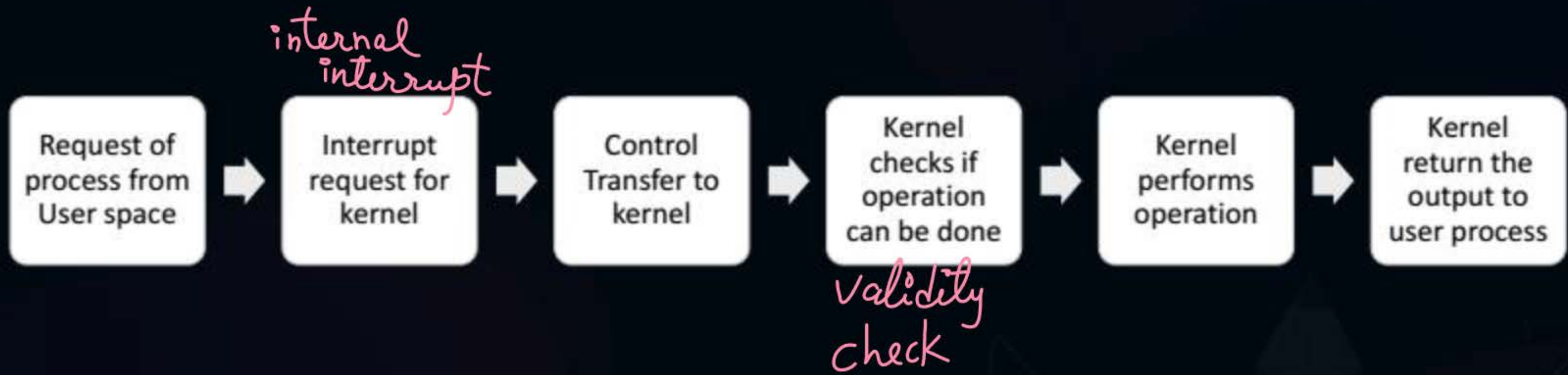


Programmatic way in which a computer program requests a service from kernel

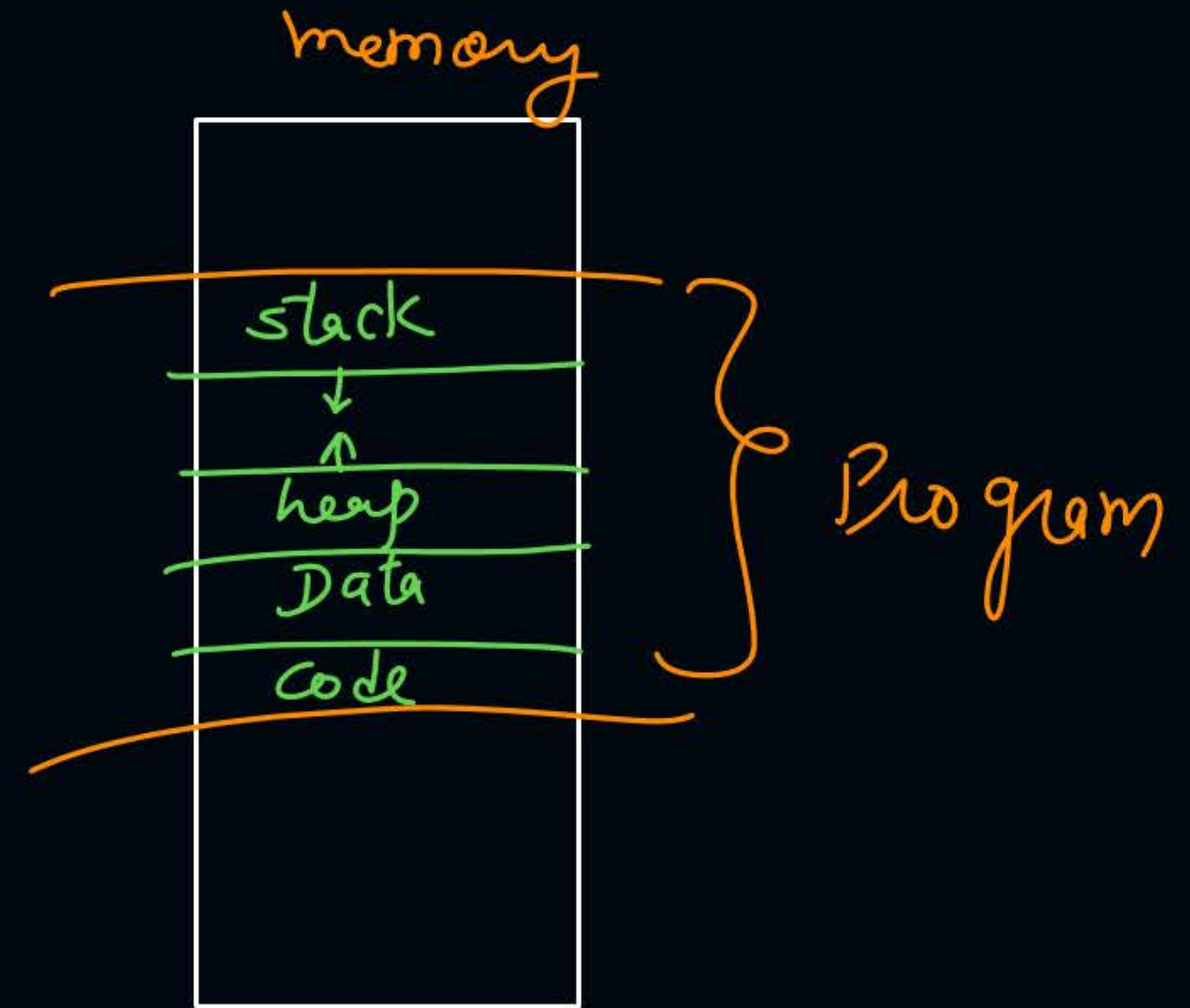




Topic : How System Call Works



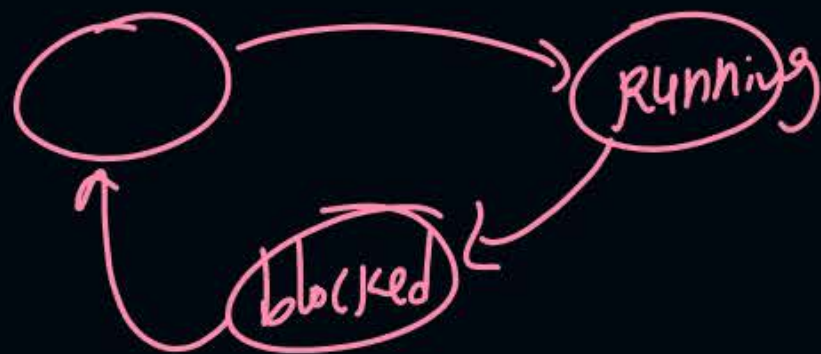
operation	system call
printf();	Yes
scanf();	Yes
malloc();	No
sqrt();	No



system call

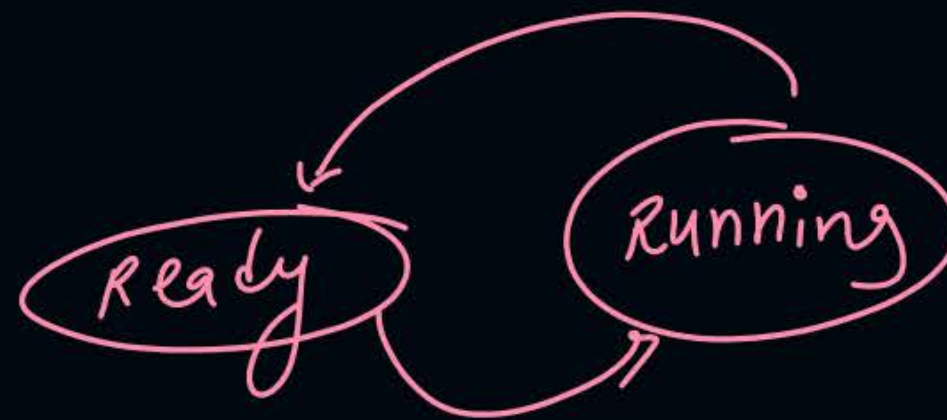
Blocking

when a process makes such system call then process will be blocked until system call execution is over.



Non-blocking

Process will not be blocked



System Call

	Windows	Unix
Process Control	CreateProcess() ExitProcess() WaitForSingleObject()	fork() exit() wait()
File Manipulation	CreateFile() ReadFile() WriteFile() CloseHandle()	open() read() write() close()
Device Manipulation	SetConsoleMode() ReadConsole() WriteConsole()	ioctl() read() write()
Information Maintenance	GetCurrentProcessID() SetTimer() Sleep()	getpid() alarm() sleep()
Communication	CreatePipe() CreateFileMapping() MapViewOfFile()	pipe() shmget() mmap()
Protection	SetFileSecurity() InitializeSecurityDescriptor() SetSecurityDescriptorGroup()	chmod() umask() chown()



2 mins Summary

Topic

Multithreading

Topic

System Call: Fork()



Happy Learning

THANK - YOU

